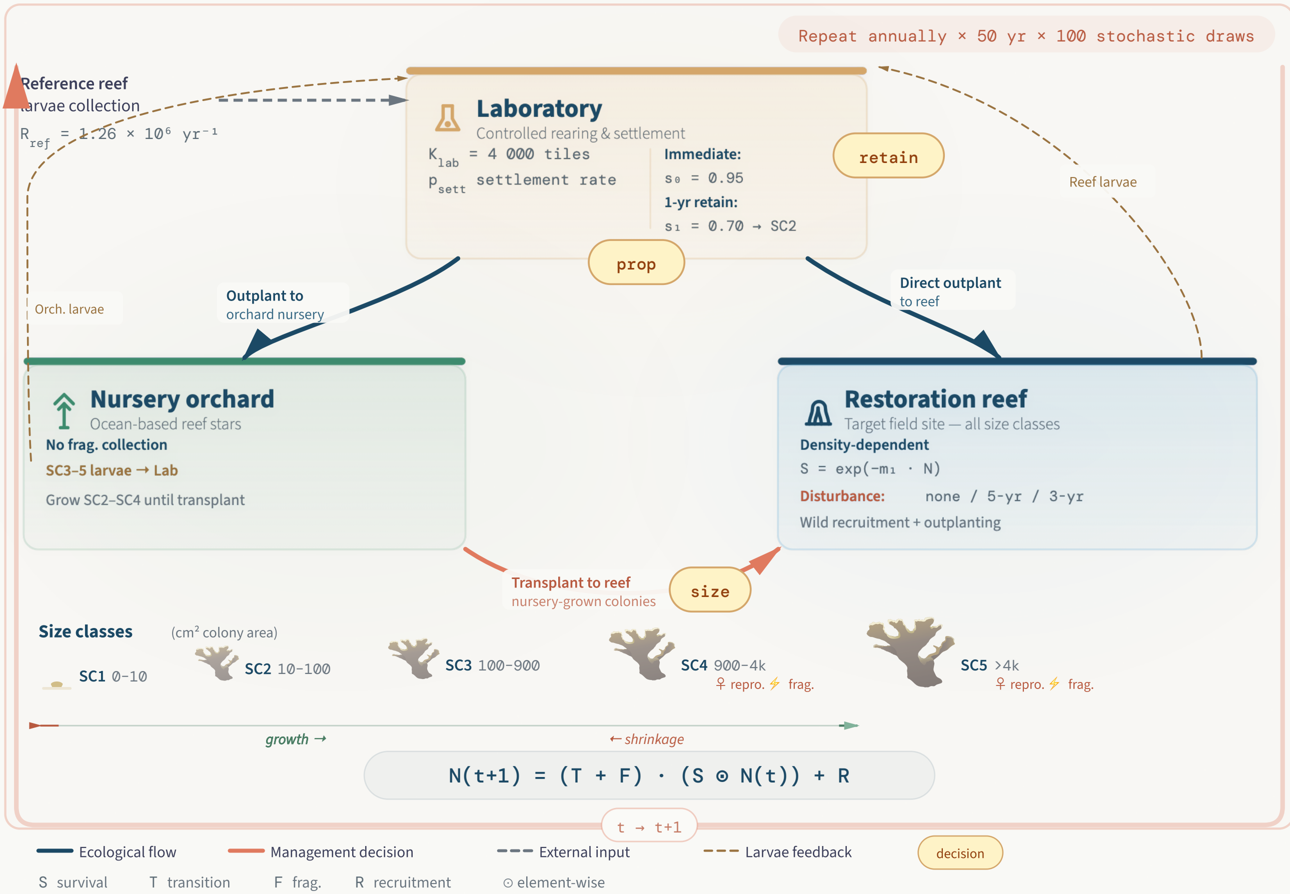

Repeat annually × 50 yr × 100 stochastic draws
Reference reef
larvae collection
R_ref = 1.26 × 10⁶ yr⁻¹
Laboratory
Controlled rearing & settlement
K_lab = 4 000 tiles
p_sett settlement rate
Immediate:
s₀ = 0.95
1-yr retain:
s₁ = 0.70 → SC2
retain
prop
Reef larvae
Orch. larvae
Outplant to
orchard nursery
Direct outplant
to reef
Nursery orchard
Ocean-based reef stars
No frag. collection
SC3–5 larvae → Lab
Grow SC2–SC4 until transplant
Restoration reef
Target field site — all size classes
Density-dependent
S = exp(−m₁ · N)
Disturbance: none / 5-yr / 3-yr
Wild recruitment + outplanting
Transplant to reef
nursery-grown colonies
size
Size classes (cm² colony area)
SC1 0–10
SC2 10–100
SC3 100–900
SC4 900–4k
♀ repro. ⚡ frag.
SC5 >4k
♀ repro. ⚡ frag.
growth →
← shrinkage
N(t+1) = (T + F) · (S ⊙ N(t)) + R
t → t+1
Ecological flow
Management decision
External input
Larvae feedback
decision
S survival
T transition
F frag.
R recruitment
⊙ element-wise